When Is a Single SHAP Run Enough? Repeat-Run Reliability of Sampled Shapley Attributions on Tabular Models

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Abstract

SHAP explanations are usually produced by a randomized estimator, run once per instance under a fixed compute budget, and then read as "the reason" for a prediction. We ask when that single run can be trusted. Exact Shapley values are available as ground truth for 8–20 players if all 2^d coalitions are tabulated, so we can repeat each estimator 100 times at budgets of 64–8192 coalition evaluations on 1,536 explained instance–model pairs (5 public tabular datasets, logistic regression, MLP, random forest, XGBoost), and check against a real-library run on 4 of these datasets and a d=262 stress test. The answer depends on the claim and the budget. At a budget close to the `shap` default (m=2048), one run names the correct top-1 feature in 99.4% of cases and the correct top-3 set in 98.4% (cell-weighted means, $d \geq 16$), but 6% of instance–model pairs still fail a 90%-repeatability criterion at m=2048 and 21% at m=256; every failure at m=2048 is a near-tie (3rd vs. 4th attribution within 2% of $\|\phi\|_2$). Reliability collapses onto one curve, $\Phi(z)$, of an instance's gap-to-noise ratio z. z can be estimated from the run itself with a sandwich standard error at no extra model calls: certifying runs with $z \geq 2$ keeps 92% of runs at m=2048 with 0.04% top-3 errors, and has lower error than a cost-matched two-run agreement check at equal coverage for $m \leq 512$. Coalition-sampling noise is much smaller than the variation from retraining the model or redrawing a 100-row background; at fixed model-call budget, background rows are the better use of compute for $d \leq 10$. Finally, `shap`'s default `l1_reg` makes errors $10\times$ larger at d=20 and fails at d=262.

1 Introduction

Shapley-value attributions [Lundberg and Lee, 2017] are used to justify individual predictions in credit, health and other high-stakes settings. For anything other than trees or linear models, they are estimated by sampling coalitions of features, and the quality of the estimate is set by a compute budget: every coalition costs one pass of the model over the whole background sample. The common practice is to run the explainer once per instance and report the resulting attribution vector. Whether this is sufficient has two possible failure modes. The estimator may be noisy, so that a second run would name different features; and the noise, or the budget, may be unimportant next to other unavoidable sources of variation in the explanation, so that worrying about it is wasted effort. Both are empirical questions, and which one holds depends on the *claim* made ("feature j matters most", "these three features", "feature j contributes $+0.07$"), the budget, the model, and the instance.

We study the question with a design that makes it checkable. For $d \leq 20$ players we tabulate the value function on all 2^d coalitions, so that (i) the exact Shapley values are known for the same game the estimators target, and (ii) running an estimator reduces to a table look-up that returns exactly what

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a run with real model calls would return, which lets us repeat each run 100 times at nine budgets. We validate this against the `shap` library with real model calls. Our contributions are:

- **A claim-level reliability protocol** with exact ground truth, repeat runs at explicit budgets, several agreement measures (top-1, top-3 set, RBO [Webber et al., 2010], Nogueira stability [Nogueira et al., 2018]) and an instance-level criterion, with cluster-bootstrap uncertainty (§3).
- **Empirical answers** over 5 datasets $\times$ 4 models $\times$ 3 estimators (§4.1–4.3): single runs are adequate for coarse claims at moderate budgets, not for finer claims at low budgets or near-ties, and the reliability of the top-3 claim is a single function of one noise-to-gap statistic.
- **A single-run certificate** from a within-run standard error that costs no extra model calls, evaluated against exact truth and against a cost-matched repeat check (§4.4).
- **Yardsticks**: the estimator's disagreement is compared to that caused by retraining the model, by redrawing the background, and by the defaults of the `shap` library (§4.5, 4.6).

We report negative and null findings as they came out (e.g. the leverage-score sampler of Musco and Witter [2024] gives no meaningful gain over Kernel SHAP at our budgets; the high-dimensional stress test is easier than anticipated for our estimators, though not for the library default).

2 Related work

Shapley estimators. Kernel SHAP [Lundberg and Lee, 2017] casts Shapley estimation as constrained weighted regression over sampled coalitions; permutation/Monte-Carlo sampling goes back to Štrumbelj and Kononenko [2014]; TreeSHAP [Lundberg et al., 2020] is exact and deterministic for trees. Covert and Lee [2021] analyze Kernel SHAP, propose an unbiased variant, variance reduction, and use bootstrap-type estimates for convergence detection and uncertainty; Mitchell et al. [2021] improve permutation sampling; Musco and Witter [2024] propose Leverage SHAP with provable guarantees; Chen et al. [2023] survey the algorithms. These works measure estimator error on small numbers of models or instances; we instead ask what that error means for explanation *claims* across many instances and models, and what a practitioner can do within a single run. We adopt the within-run uncertainty idea of Covert and Lee [2021] and test it (in a sandwich-variance form for the paired, partially enumerated sampler used in practice) as a claim-level certificate.

Variability of explanations. Krishna et al. [2024] study disagreement between different explanation methods; Yuan et al. [2022] study how the random background sample changes SHAP values and rankings for deep models on one clinical dataset; Chen et al. [2020] discuss how the choice of expectation (interventional vs. observational) changes what Shapley values mean. We use the interventional game with a fixed background and then compare its estimator noise to the background and model effects. Our retraining comparison is in the spirit of predictive multiplicity [Marx et al., 2019]: equally accurate models need not give the same explanation.

Measures. We use stability of top-k selection [Nogueira et al., 2018] and rank-biased overlap [Webber et al., 2010], and the datasets of Ding et al. [2021] and the UCI/OpenML repositories.

3 Setting and methodology

Target and budget. For a model f (probability of the positive class), an instance x and a background sample $B = \{b_1..b_n\}$ (n=100 rows unless stated), the interventional value function is $v(S) = \frac{1}{n}\sum_r f(x_S, b_{r,\bar{S}})$ over d players (original features; one-hot columns of a categorical feature form one player). The Shapley values $\phi \in \mathbb{R}^d$ of v are the *target*; we call them exact. A *single run* of an estimator with budget m evaluates m coalitions (excluding $\emptyset$ and the full set) and returns $\hat{\phi}$; its cost is $m \cdot n$ model calls. Randomness is only in the coalitions (background fixed) unless stated.

Estimators. *Kernel SHAP*: our numba port of the sampling logic of `shap.KernelExplainer` (paired sampling of complements, complete enumeration of the cheapest size classes while the budget allows, size-proportional sampling of the rest with duplicate counting, constrained weighted least squares) without the L1 feature selection step. *Leverage SHAP*: the same machinery, drawing

Dataset	n	n_{train}	d	b_{ref}	#inst.	test AUC			
						LR	MLP	RF	XGB
Pima Diabetes	768	537	8	500	100	0.867	0.869	0.847	0.834
Give Me Some Credit	40000	28000	10	1000	100	0.853	0.864	0.865	0.866
ACSIncome (CA 2018)	40000	28000	10	1000	100	0.889	0.899	0.898	0.902
ACSEmployment (CA 2018)	40000	28000	16	100	60	0.876	0.902	0.904	0.905
German Credit	1000	700	20	100	24	0.798	0.778	0.802	0.785

Table 1: Datasets and models. d: players; b_{ref}: size of the reference background sample used to tabulate v (the main experiments use its first 100 rows); #inst.: explained test instances per model.

coalition sizes uniformly (the leverage scores of the Shapley regression are $1/\binom{d}{|S|}$, which we verified numerically) with Horvitz–Thompson weights; this is our re-implementation of the idea in Musco and Witter [2024], not their code. *Permutation SHAP*: antithetic (forward and reversed) permutations, $2(d-1)$ coalitions per pair. Details are in Appendix A.

Ground truth and the table oracle. For each instance we evaluate v on all 2^d coalitions ($d \le 20$), compute ϕ exactly, and implement every estimator as a look-up in this table. Since the estimators only access v through the values on the coalitions they sample, this returns exactly what a run with real model calls would return. We verify the equivalence against the real `shap` library in §4.3. Budgets are $m \in \{32, 64, \ldots, 8192\}$; $m = 32$ is below $2d$ for $d \ge 16$, where the regression is under-determined and a run can be arbitrarily wrong (we report it in the data release but exclude it from all analyses; Permutation SHAP needs $m \ge 2(d-1)$). A dataset with $2^d - 2 \le m$ is enumerated exactly. The `shap` default is $m = 2d + 2048$.

Data, models, instances. Table 1 lists the datasets: Pima Diabetes (OpenML 37), Give Me Some Credit (OpenML 46929, 40k-row subsample), ACSIncome and ACSEmployment [Ding et al., 2021] (California, 2018, 40k-row subsample), and German Credit [Hofmann, 1994]. Models: ℓ_2 logistic regression, a 2-layer MLP, random forest [Breiman, 2001] and XGBoost [Chen and Guestrin, 2016] (scikit-learn [Pedregosa et al., 2011]; hyper-parameters fixed a priori, Appendix B). One model per (dataset, model type) on a stratified 70/30 split; instances are drawn uniformly from the test split (100 each for $d \le 10$, 60 for d=16, 24 for d=20 where each instance costs $2^{20} \cdot 100$ model calls to tabulate). For each (instance, estimator, m) we do 100 independent runs. A replicate with a second split and model seed is in Appendix D.

Claims and reliability measures. For each run we compare $\hat{\phi}$ to ϕ on: relative ℓ_2 error; **top-1** (is $\arg\max_j |\hat{\phi}_j|$ the exact one); **top-3 set** (is the set of the three largest $|\hat{\phi}_j|$ exactly the exact set); sign agreement on the exact top-3; Spearman correlation and RBO (p=0.9) of the $|\cdot|$-rankings. For repeatability we also compare 200 random pairs of independent runs (agreement of top-3 sets, RBO) and compute the Nogueira stability of top-3 selection over the 100 runs. The **instance-level criterion** is: a single run is *reliable for a claim* on instance x if the claim is correct in at least 90% of runs; we report the fraction of instances that satisfy it (so a run-averaged success rate of 98% can hide a 10% minority of unreliable instances). The 0.9 level is a convention, not a standard; Fig. 1 and Appendix C show other views.

Statistics. The unit of analysis is the explained instance. Results per (dataset, model) cell are means over instances; pooled numbers are equal-weight means over cells; 95% intervals are percentile intervals from a cluster bootstrap that resamples instances within each cell (1000–2000 resamples). They quantify *instance sampling* uncertainty and, via the 100 runs per instance, run-to-run noise; they do not include variation across independently trained models, which we address only by the small replicate and the retraining experiment. Intervals for rates near 0 (e.g. certified error) are bootstrap intervals over instance clusters and are approximate.

Single-run diagnostic. For Kernel SHAP the sampled draws $j = 1..N$ (a draw is a coalition and its complement) are iid, so the estimator $\hat{\phi} = A^{-1} \sum_j g_j$ in the reduced $(d-1)$-parameter regression has a sandwich covariance $\hat{V} = A^{-1}\big[\frac{N}{N-1} \sum_j (g_j - \bar{g})(g_j - \bar{g})^\top\big] A^{-1}$ in which the enumerated part

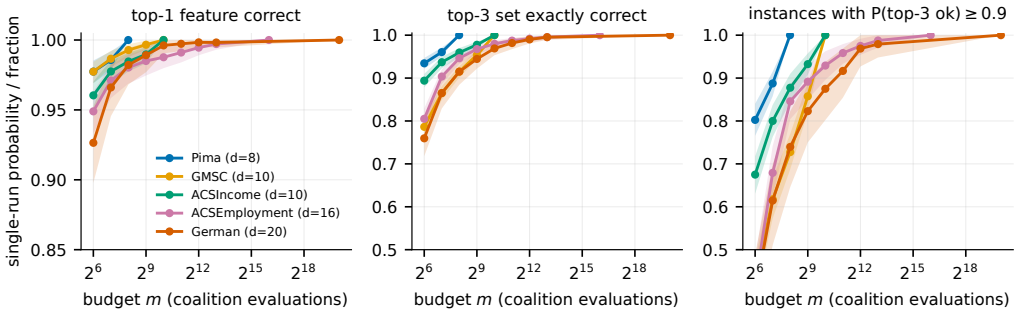


Figure 1: Single-run reliability of Kernel SHAP vs. budget for the five datasets (mean over 4 models; bands: 95% cluster-bootstrap CI). The final point of each line is complete enumeration ($2^d - 2$ evaluations), which is exact. Left: probability that the top-1 feature is correct; middle: that the top-3 set is correct; right: fraction of instances for which the top-3 claim is correct in $\geq$90% of runs.

contributes no variance; for Permutation SHAP the standard error is the sample SD over permutation pairs divided by $\sqrt{\#\text{pairs}}$. Neither needs a model call beyond the run. For the top-3 claim, sort $|\hat{\phi}|$ and define $z_3 = (a_{(3)} - a_{(4)})/\sqrt{\widehat{\text{se}}^2_{(3)} + \widehat{\text{se}}^2_{(4)}}$, and the *certificate* "$z_3 \geq c$"; the *oracle* version z uses the exact gap and the empirical SD of the 100 runs.

4 Results

4.1 How reliable is one run? It depends on the claim and the budget

Fig. 1 and Table 5 (Appendix) give the main picture. For the two datasets with $d \geq 16$, pooled over the 8 (dataset, model) cells, a Kernel SHAP run at m=2048 (the default budget is m=2d+2048) names the exact top-1 feature in 99.4% [99.0, 99.8] of runs and the exact top-3 set in 98.4% [97.7, 99.1], with mean relative ℓ_2 error 1.4%. The top-3 claim is considerably weaker at lower budgets: 93.1% [91.4, 94.6] at m=256 and 78.2% [75.9, 80.8] at m=64 (top-1: 98.1% and 93.8%). The same budget yields different instance-level guarantees: the fraction of instances for which the top-3 claim meets the 90% criterion is 93.8% [90.7, 96.6] at m=2048, 79.3% [74.6, 83.9] at m=256 and only 40.6% [35.2, 46.5] at m=64. Among the 336 instance–model pairs with $d \geq 16$ the median budget needed to meet the criterion is 128, the 90th percentile 1024, 5.4% need more than 2048 and 1.5% do not reach it by 8192. Run-to-run agreement of the top-3 set between two independent runs is 97.6% [96.5, 98.5] at m=2048 and 90.1% [88.0, 92.1] at m=256 (Nogueira stability 0.991 and 0.960; this measure is lenient here because only two or three features fluctuate between ranks).

Three qualifications matter. (i) For $d \leq 10$ the problem disappears at no larger cost than the default: complete enumeration needs $2^d - 2 \leq 1022$ evaluations, so noise can be eliminated; at the intermediate m=256 the top-3 claim is correct in 86–98% of runs (GMSC) and 93–97% (ACSIncome) depending on the model (Table 5). (ii) Models differ, but not in a stable order: at m=2048 the fraction of reliable instances for $d \geq 16$ is 96% for LR, 96% for RF, 94% for XGBoost and 92% for MLP (24–60 instances each), and a replicate with a different training split moves single cells by up to 16 points (Appendix D), so we do not rank model families. (iii) Top-1 is much easier than top-3, and values are easier than ranks near ties: the pooled mean relative error is 4.8% at m=256 and 0.6% at m=8192.

4.2 Where and why single runs fail

A run errs on the top-3 set when noise reorders the 3rd and 4th largest attributions. If the estimate were Gaussian with the empirical SD σ_j of each feature, the probability of preserving the order is $\approx \Phi(z)$ with $z = (a_{(3)} - a_{(4)})/\sqrt{\sigma^2_{(3)} + \sigma^2_{(4)}}$. Fig. 2a shows this holds empirically: instance-by-budget cells from all five datasets, four models and all budgets lie close to $\Phi(z)$, regardless of d (the observed probability is slightly below $\Phi(z)$ for $z \lesssim 2$, e.g. 0.54 vs. 0.60 at $z \approx 0.25$: the

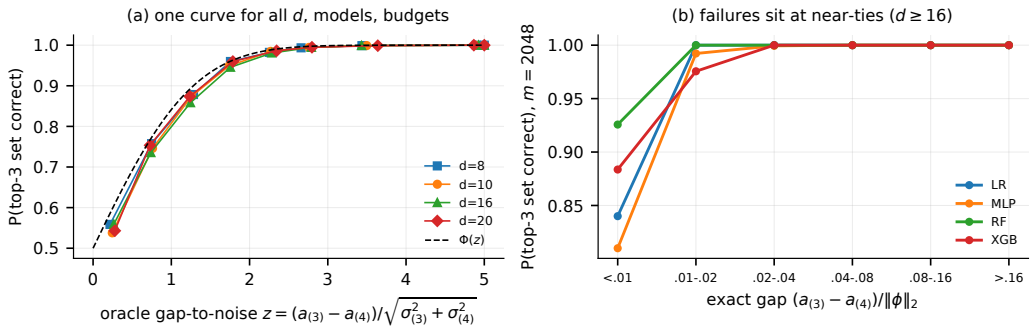


Figure 2: (a) Probability that a run recovers the exact top-3 set versus the oracle gap-to-noise ratio z (binned; Kernel SHAP, all datasets, models and budgets $m \geq 64$, bins with ≥ 20 instance–budget cells). The dashed line is the standard normal CDF. (b) At $m = 2048$ and $d \geq 16$ the failures are concentrated at near-ties of the exact attribution.

m	P(top-3 set correct), %			median rel. ℓ_2 error, %		
	Kernel	Leverage	Perm.	Kernel	Leverage	Perm.
64	78.2	79.4	81.1	12.25	11.40	7.57
128	88.4	88.8	87.5	6.54	6.22	5.08
256	93.1	93.5	90.9	4.00	3.73	3.58
512	95.6	96.2	94.0	2.60	2.43	2.44
2048	98.4	98.4	97.5	1.13	1.08	1.21
8192	99.5	99.6	99.2	0.47	0.46	0.61

Table 2: Estimators at equal budget, $d \geq 16$ (336 instance–model pairs): pooled P(top-3 set correct) and median relative ℓ_2 error. Paired differences in top-3 correctness (Leverage−Kernel, Permutation−Kernel, 95% CI): m=64: +1.1 [0.4, 1.8] / +2.9 [1.9, 3.9] points; m=256: +0.4 [−0.1, 0.9] / −2.2 [−2.9, −1.4]; m=2048: −0.0 [−0.2, 0.1] / −0.9 [−1.3, −0.6].

Gaussian two-feature approximation is mildly optimistic). The practical consequence is that the budget needed is set by the *instance's* margin: the gap between the 3rd and 4th attribution relative to the run-to-run noise, where the noise shrinks faster than $m^{-1/2}$ (by a factor of about 1.5 per doubling of m from m=256, since larger budgets also enumerate more size classes; Appendix Fig. 6). At m=2048, 19.6% of the $d \geq 16$ instance–model pairs have a gap below $0.02\|\phi\|_2$, and *all 18* pairs that fail the 90% criterion are among them (Fig. 2b): failures are near-ties. In those cases the claim "these are the top three features" is not well determined by the model's exact attribution either, so resolving them with more compute has little value. The noise level itself is tied to non-additivity: the share of the kernel-weighted variance of v left unexplained by the additive game correlates with the relative run-to-run SD (Spearman 0.64 at m=512, $d \geq 16$, 336 instance–model pairs; 0.70 within ACSEmployment, 0.51 within German).

4.3 Estimators, and what the library default does

The three estimators are close (Table 2). Leverage SHAP differs from Kernel SHAP by at most +1.1 points of top-3 correctness at any budget (largest at m=64) and is indistinguishable from m=2048 on; Permutation SHAP is more accurate at the lowest budget but 0.4–2.2 points worse for top-3 from m=128 on. So the estimator is not the lever: the conclusions of §4.1–4.2 are not artefacts of Kernel SHAP.

Validation. Running the real library (`shap` 0.51.0, XGBoost, 8 instances × 20 repeats per setting, real model calls) gives errors that match the table oracle at equal budget for Kernel SHAP with `l1_reg=False`, e.g. median relative error on German 0.195/0.061/0.018 (library) vs. 0.217/0.067/0.020 (oracle) at m=64/256/2088 (Appendix Table 7). The library's permutation explainer is somewhat worse than ours on ACSEmployment at m=64 (0.090 vs. 0.053; 8 instances) and similar elsewhere. *The default matters.* The library's default is `l1_reg="num_features(10)"`,

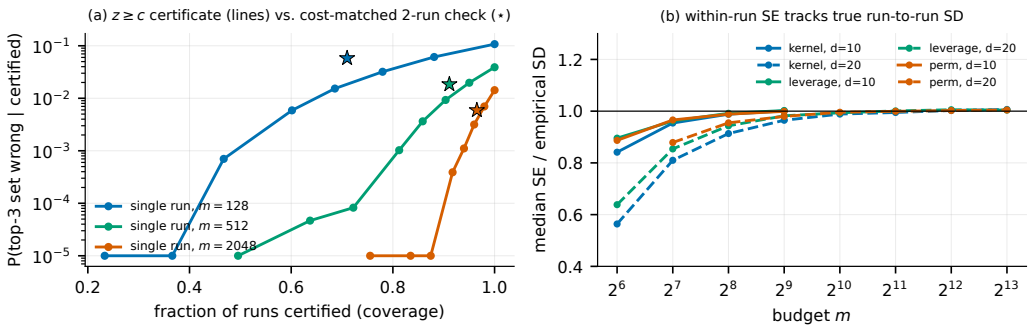


Figure 3: (a) Error of the top-3 set among certified runs vs. certified fraction, as the threshold c in "$z_3 \geq c$" varies (lines; $d \geq 16$ datasets, Kernel SHAP), and the cost-matched alternative of running twice at $m/2$ and certifying when the two top-3 sets agree (stars, error of the pooled estimate). (b) Ratio of the within-run standard error to the empirical SD across runs.

m	all runs: err. (%)	AUROC(z_3)	$z_3 \geq 2$: cov. / err. (%)	2-run agree: cov. / err. (%)	z_3 at 2-run coverage: err. (%)
128	10.75	0.897	0.60 / 0.59 [0.42, 0.79]	0.71 / 5.80 [4.74, 6.93]	1.87 [1.44, 2.35]
512	3.90	0.957	0.81 / 0.10 [0.05, 0.18]	0.91 / 1.85 [1.33, 2.44]	1.07 [0.72, 1.50]
2048	1.43	0.982	0.92 / 0.04 [0.00, 0.08]	0.97 / 0.59 [0.32, 0.91]	0.39 [0.20, 0.63]

Table 3: Single-run certificate vs. two half-budget runs ($d \geq 16$; clusters = instance–model pairs; 95% cluster-bootstrap CIs). "All runs": top-3 error without any certification. cov.: fraction certified.

which keeps at most 10 features. For $d > 10$ its error stops decreasing with the budget: German, default budget: 0.210 median relative error vs. 0.018 without it (top-3 set 98.8% vs. 100%); ACSEmployment: 0.031 vs. 0.012. This is a property of that default in that version, not of Kernel SHAP; it is invisible if one only repeats runs, since the error is a bias.

4.4 Certifying a single run from itself

Because the SE is computed from the sampled coalitions of the run itself, certification costs no model calls. The within-run SE tracks the empirical run-to-run SD (Fig. 3b: median ratio ≥ 0.96 from m=512 on at $d \geq 16$; it underestimates by 32–44% at m=64, 15–19% at m=128 and 5–9% at m=256). The statistic z_3 discriminates correct from incorrect runs (AUROC 0.90, 0.96, 0.98 at m=128, 512, 2048). Certifying runs with $z_3 \geq 2$ leaves an error of 0.04% [0.00, 0.08] at m=2048 (cov. 92%), 0.10% [0.05, 0.18] at m=512 (cov. 81%) and 0.59% [0.42, 0.79] at m=128 (cov. 60%), against uncertified errors of 1.43%, 3.90% and 10.75%. The cost-matched alternative (two independent runs at $m/2$, certify if the top-3 sets agree) is worse at equal coverage: at the coverage of the agreement check, thresholding z_3 gives 1.87% [1.44, 2.35] vs. 5.80% [4.74, 6.93] error (m=128) and 1.07% vs. 1.85% (m=512); at m=2048 the difference (0.39% vs. 0.59%) is within noise. Uncertified runs are not necessarily wrong (at m=2048 about 72% of uncertified runs, $z_3 < 1$, are still correct) but their top-3 claim should be reported with a warning, a larger budget, or as "tied".

4.5 Yardsticks: retraining, background, and a fixed model-call budget

Is estimator noise large compared to other things that change an explanation? Fig. 4 (Appendix Table 6) compares the pairwise agreement of two single runs with that of two *exact* explanations that differ only by (a) the 100 background rows, redrawn from a reference sample of 500–1000 rows, or (b) retraining on a bootstrap resample of the training set with a new seed (the test AUC of the 8 models per cell spans at most 0.04). At $m = 128$ two runs agree on the top-3 set in 81–95% of cases (dataset means over the 4 models), whereas two backgrounds agree in 53–77% and two retrained models in 32–66%; for relative ℓ_2 differences the order is the same (0.03–0.06 vs. 0.14–0.29 vs. 0.23–0.44). For $d \leq 10$ there is thus no case for spending compute on more repeated runs while the background and the model remain as variable as they are; conversely, redrawing a 100-row background between runs adds far more variation than coalition sampling. Fixed-budget view (Fig. 5): with total budget

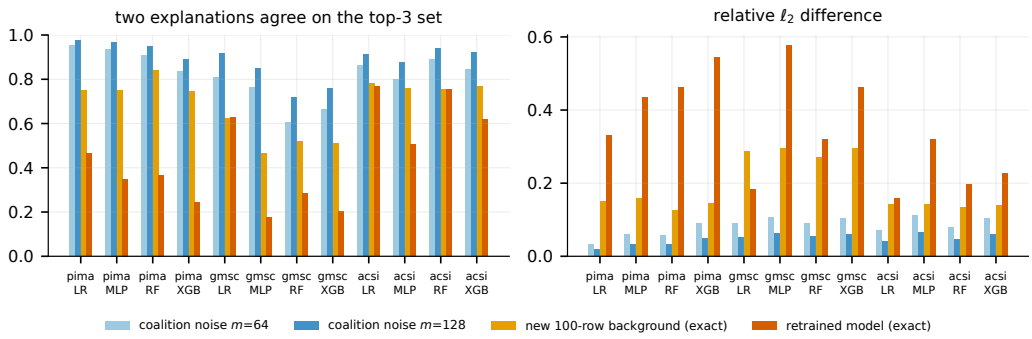


Figure 4: Agreement between two explanations of the same instance that differ in one source only. Coalition-sampling noise of a single run (Kernel SHAP, m=64, 128) vs. two exact explanations from independently drawn 100-row backgrounds or from two bootstrap-retrained models (K=8 models; AUC range at most 0.04 per cell). Datasets with $d \leq 10$, 50 instances per cell. Left: same top-3 set. Right: relative ℓ_2 difference.

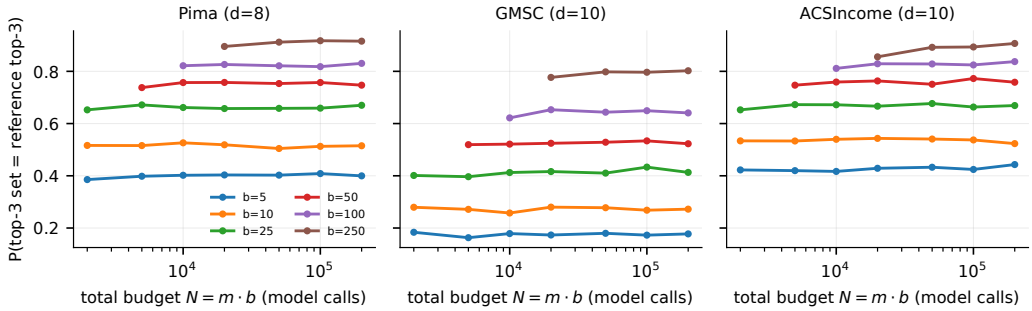


Figure 5: Fixed model-call budget $N = m\,b$ split between coalitions m and background rows b (Kernel SHAP, fresh random background per run, 4 models, 50 instances, 20 runs). Only $m \geq 64$ is shown; where $2^d - 2 < N/b$ all coalitions are enumerated.

$N = m \cdot b$ model calls, reliability against the 1000-row reference explanation is nearly flat in m once $m \geq 64$ and increases with b (ACSIncome, $N = 2 \cdot 10^4$–10^5: top-3 set correct 0.66–0.68 at b=25, 0.82–0.83 at b=100, 0.86–0.89 at b=250; GMSC: 0.41–0.43, 0.64–0.65, 0.78–0.80); the best use of a fixed budget is to enlarge the background and keep m at a few hundred. (Because the reference itself is a finite sample, the advantage of large b is partly an overlap effect; the monotone trend, not the absolute level, is the evidence.)

4.6 A high-dimensional stress test

To probe d far beyond what can be tabulated we use UCI Arrhythmia [Guvenir et al., 1997] (d=262 non-constant features, 316 training rows, XGBoost, test AUC 0.870, log-odds output), where the exact interventional TreeSHAP values with the same 100-row background are the ground truth (additivity error $\leq 10^{-6}$), 16 instances, 20 runs per setting, real model calls. Table 4 shows that, for this model, our Kernel SHAP is accurate at the library budget $m \approx 2d+2048$: top-1 always correct and top-3 correct in 99% (95% CI over instances 0.97–1.00) at $m = 2048$ (the library with `l1_reg=False`: 95% [0.86, 1.00] at 2572 coalitions, so the two overlap within the small-sample noise) ($m < 2d$ is unusable: top-3 correct in only 73% at $m = 512$). This is not a general statement about high dimensions: for this model the 10 largest attributions carry on average 74% of $\|\phi\|_2^2$ (minimum 49%), so most players are nearly irrelevant. The library default, however, collapses: median relative error 0.62, top-1 correct 65%, top-3 41%, because LARS keeps 10 of 262 features.

Estimator	m	median rel. ℓ_2 err.	top-1 correct	top-3 set correct
Kernel SHAP (ours, no L1)	1024	0.051	0.997 [0.99,1.00]	0.962 [0.90,1.00]
Kernel SHAP (ours, no L1)	2048	0.031	1.000 [1.00,1.00]	0.991 [0.97,1.00]
Kernel SHAP (ours, no L1)	8192	0.013	1.000 [1.00,1.00]	0.988 [0.97,1.00]
Leverage SHAP (ours)	1024	0.037	1.000 [1.00,1.00]	0.947 [0.88,1.00]
Leverage SHAP (ours)	2048	0.020	0.997 [0.99,1.00]	0.963 [0.91,1.00]
Leverage SHAP (ours)	8192	0.009	1.000 [1.00,1.00]	0.975 [0.93,1.00]
shap.KernelExplainer, l1_reg=False	2572	0.021	1.000 [1.00,1.00]	0.950 [0.86,1.00]
shap.KernelExplainer, default	2572	0.623	0.650 [0.42,0.86]	0.412 [0.21,0.64]

Table 4: Arrhythmia (d=262): comparison with exact TreeSHAP; 16 instances, mean over instances with 95% bootstrap CI.

5 Discussion and limitations

Verdict. The answer to "can a single run be taken as the basis of the prediction?" is conditional. For the *top-1* claim, yes at almost any budget $\geq$256 on our data (98%+ correct). For the *top-3 set*, yes at $m \approx 2048$ for most instances (98.4% of runs; 94% of instances meet a 90% repeatability bar) but not at $m \leq 256$ for a fifth of instances, and not for the near-tie instances (about one in five at $d \geq 16$) at any budget—there the question is ill-posed rather than under-sampled. For complete enumeration ($d \lesssim 10$) there is no noise to worry about. A single run is also adequate in a more relative sense: coalition noise is much smaller than the changes due to retraining or the background draw. What fails is not the single run itself but unverified single runs: a free within-run z statistic identifies the runs that cannot be trusted, and the `shap` defaults may bias the result in a way no repetition would reveal.

Limitations. (1) *Scale:* $d \leq 20$ for tabulated ground truth and one stress test at d=262 with trees; non-additive high-dimensional games are not covered, so the budget requirements we measure should not be extrapolated to e.g. $d \geq 50$ with strong interactions. (2) *Value function:* interventional with a fixed 100-row background, probability output (log-odds for the stress test); conditional/observational variants [Chen et al., 2020] and other outputs were not studied. (3) *Models and instances:* four standard model families, one trained model per cell (plus a replicate and the retraining experiment), 24 instances for d=20; CIs reflect instance sampling, not model-training randomness; the replicate (Appendix D) shows cell-level differences of up to 16 points in top-3 correctness, so absolute levels are model-specific, and the German results are the least precise. Tree models have an exact algorithm [Lundberg et al., 2020] and our findings concern model-agnostic use. (4) *Criteria:* the 0.9 level, top-3 and exact-set claims are conventions; other claims (e.g. values within 10%, sign) give different budgets. (5) *Estimators:* our Kernel SHAP is a port of the library's sampling logic, validated against the library but not identical (duplicate handling); Leverage SHAP is our re-implementation (without-replacement schemes and the authors' tuning are not included), so its null result is not a statement about the original. (6) The background experiment uses a finite reference sample; its absolute levels are optimistic for large b. (7) The certificate's SE assumes iid coalition draws and underestimates at d=20 and $m \leq 128$; we evaluated one certificate form, and the calibration at tail error rates has wide intervals.

Conclusion. Single-run SHAP is reliable for coarse claims at moderate budgets, unreliable for fine rankings at low budgets or at near-ties, and its largest practical risks are the background and library defaults rather than coalition noise. We recommend reporting the budget, using enumeration for $d \lesssim 10$, using the within-run z to flag unresolvable rankings, and checking default settings such as `l1_reg`.

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A Estimator details

Let $k(S) \propto 1/(\binom{d}{|S|}|S|(d-|S|))$ be the Shapley kernel. *Kernel SHAP.* Size classes $s = 1..\lceil (d-1)/2 \rceil$ stand for sizes s and $d-s$ (paired). Starting from the cheapest class, a class is enumerated completely (with complements) if its share of the remaining budget covers it; otherwise sizes are drawn with probability proportional to the kernel mass of the remaining classes, a random coalition of that size is drawn and its complement added; a repeated coalition increments the weight of the existing row; sampling stops when m rows exist. Sampled rows jointly receive the kernel mass left after enumeration, in proportion to counts; the regression with the sum-to-effect constraint is solved by elimination of the last player (as in `shap`). *Leverage SHAP*: identical, but the size class is drawn uniformly over the remaining classes and each draw contributes weight $k(S)\binom{d}{|S|}/(\pi_s N)$ (π_s the class probability, N the number of draws). The leverage scores of the rows of the constrained design $\sqrt{k(S)}Z_S$ equal $1/\binom{d}{|S|}$ (checked numerically for $d = 7$). *Permutation SHAP*: $\lfloor m/(2(d-1)) \rfloor$ forward/reverse permutation pairs. Unbiasedness and SE calibration were checked on random games with $d = 10, 20$ (code `test_est.py`).

B Setup details

Models: LR (C=1); MLP $64-64$ ReLU, Adam, $\alpha = 10^{-3}$ with early stopping for $n_{\text{train}} \geq 5000$, $\alpha = 1$, 50 epochs without early stopping for smaller data (chosen from 5 splits of German/Pima to avoid the overfitting seen with early stopping on 10% validation); RF 100 trees (50 if $n_{\text{train}} < 1000$), min leaf 5 (20 if $n \geq 5000$), $\sqrt{d}$ features; XGBoost 150 trees, depth 4, learning rate 0.1, subsample 0.8, column sample 0.8. Preprocessing is fit on the training split: median imputation; quantile-normal transform of numeric features and one-hot of categoricals for LR/MLP, ordinal codes for trees; high-cardinality ACS features (OCCP, POBP) use smoothed target means. Background rows: b_{ref} random training rows; the main experiment uses the first 100. German credit contains protected attributes (sex, age); we use it only as a benchmark. Pima zeros are not treated as missing. Seeds: instance sampling 123, background 1, split 0 (replicate: 1), per-run seeds from a hash of (dataset, model, instance, m, estimator, run).

C Further results

		$m = 256$			$m = 2048$		
Dataset	Model	top-1	top-3 set	% inst. $\geq$0.9	top-1	top-3 set	% inst. $\geq$0.9
Pima (d=8)	LR	enum.	enum.	enum.	enum.	enum.	enum.
Pima (d=8)	MLP	enum.	enum.	enum.	enum.	enum.	enum.
Pima (d=8)	RF	enum.	enum.	enum.	enum.	enum.	enum.
Pima (d=8)	XGB	enum.	enum.	enum.	enum.	enum.	enum.
GMSC (d=10)	LR	99.7	98.3	95.0	enum.	enum.	enum.
GMSC (d=10)	MLP	99.1	93.4	77.0	enum.	enum.	enum.
GMSC (d=10)	RF	99.6	86.1	56.0	enum.	enum.	enum.
GMSC (d=10)	XGB	98.7	88.3	63.0	enum.	enum.	enum.
ACSIncome (d=10)	LR	99.0	96.5	87.0	enum.	enum.	enum.
ACSIncome (d=10)	MLP	98.1	93.3	82.0	enum.	enum.	enum.
ACSIncome (d=10)	RF	98.6	97.2	92.0	enum.	enum.	enum.
ACSIncome (d=10)	XGB	98.2	96.8	90.0	enum.	enum.	enum.
ACSEmployment (d=16)	LR	99.9	97.0	93.3	100.0	99.3	96.7
ACSEmployment (d=16)	MLP	96.1	92.5	78.3	98.1	98.0	93.3
ACSEmployment (d=16)	RF	96.9	97.6	91.7	98.6	99.7	98.3
ACSEmployment (d=16)	XGB	99.3	91.5	75.0	99.8	97.9	95.0
German (d=20)	LR	100.0	95.4	87.5	100.0	98.4	95.8
German (d=20)	MLP	98.3	85.8	58.3	100.0	96.7	87.5
German (d=20)	RF	98.6	92.3	75.0	99.9	98.6	91.7
German (d=20)	XGB	96.0	92.5	75.0	99.0	99.0	91.7

Table 5: Kernel SHAP single-run reliability per dataset and model (%): top-1, top-3 set correct (means over runs and instances) and fraction of instances for which the top-3 claim is correct in $\geq$90% of runs. "enum.": $m \geq 2^d - 2$, complete enumeration (exact). CIs are in `results/summary` (code `analyze_figs.py`).

Dataset	Source of variation	top-1 agree	top-3 set agree	RBO	rel. ℓ_2 diff.
Pima	coalition sampling, m=64	0.969	0.908	0.980	0.060
Pima	coalition sampling, m=128	0.981	0.946	0.988	0.034
Pima	new 25-row background (exact)	0.827	0.558	0.927	0.317
Pima	new 100-row background (exact)	0.919	0.770	0.962	0.146
Pima	retrained model (exact)	0.743	0.356	0.888	0.444
GMSC	coalition sampling, m=64	0.966	0.711	0.960	0.098
GMSC	coalition sampling, m=128	0.980	0.811	0.975	0.057
GMSC	new 25-row background (exact)	0.655	0.302	0.851	0.603
GMSC	new 100-row background (exact)	0.811	0.530	0.919	0.287
GMSC	retrained model (exact)	0.817	0.323	0.881	0.385
ACSIncome	coalition sampling, m=64	0.946	0.851	0.964	0.093
ACSIncome	coalition sampling, m=128	0.969	0.913	0.978	0.054
ACSIncome	new 25-row background (exact)	0.838	0.565	0.921	0.291
ACSIncome	new 100-row background (exact)	0.922	0.766	0.960	0.139
ACSIncome	retrained model (exact)	0.849	0.662	0.932	0.226

Table 6: Pairwise agreement of two explanations differing in one source of variation (dataset means over 4 models, 50 instances; $d \leq 10$). Coalition sampling: two independent single Kernel SHAP runs.

Dataset	Variant	m	lib: med. rel. err.	sim: med. rel. err.	lib: top-3 ok	sim: top-3 ok
Pima	Kernel, l1_reg=False	64	0.072	0.067	0.806	0.843
Pima	Permutation	64	0.067	0.054	0.812	0.805
ACSIncome	Kernel, l1_reg=False	64	0.066	0.061	0.844	0.862
ACSIncome	Kernel, l1_reg=False	256	0.022	0.021	0.969	0.959
ACSIncome	Permutation	64	0.065	0.056	0.869	0.850
ACSIncome	Permutation	256	0.024	0.025	0.894	0.900
ACSEmployment	Kernel, l1_reg=False	64	0.124	0.122	0.812	0.847
ACSEmployment	Kernel, l1_reg=False	256	0.042	0.041	0.950	0.966
ACSEmployment	Kernel, l1_reg=False	2080	0.012	0.011	1.000	0.998
ACSEmployment	Permutation	64	0.090	0.053	0.869	0.926
ACSEmployment	Permutation	256	0.030	0.026	0.962	0.955
ACSEmployment	Kernel, default l1_reg	256	0.049	0.041	0.962	0.966
ACSEmployment	Kernel, default l1_reg	2080	0.031	0.011	1.000	0.998
German	Kernel, l1_reg=False	64	0.195	0.217	0.806	0.770
German	Kernel, l1_reg=False	256	0.061	0.067	0.956	0.963
German	Kernel, l1_reg=False	2088	0.018	0.020	1.000	1.000
German	Permutation	64	0.154	0.168	0.781	0.757
German	Permutation	256	0.073	0.066	0.925	0.939
German	Kernel, default l1_reg	256	0.223	0.067	0.875	0.963
German	Kernel, default l1_reg	2088	0.210	0.020	0.988	1.000

Table 7: Real `shap` 0.51.0 library (XGBoost, 8 instances $\times$ 20 repeats) vs. the table-oracle estimator on the same instances ("sim"; for m="auto" the library uses $2d{+}2048$, sim $m{=}2048$). Median relative ℓ_2 error and top-3 set correctness.

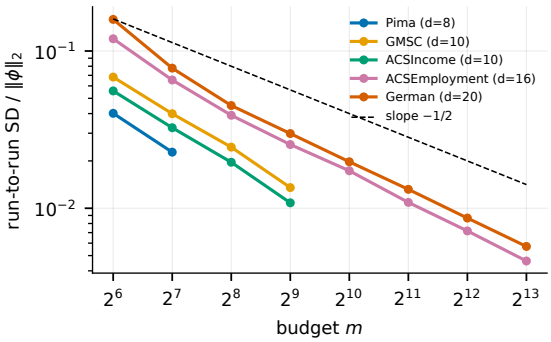


Figure 6: Run-to-run SD of the attribution vector relative to $\|\phi\|_2$ (median over instances; Kernel SHAP), by dataset; the dashed line has slope $-1/2$ for reference.

D Replicate with a second split and model seed

To see how much a different trained model changes the picture, we repeated the main experiment for the $d \leq 10$ datasets with a different split and model seed (seed 1; all else equal; 100 instances per cell). Table 8 shows top-3 correctness at two budgets. The mean absolute difference between replicates over the 12 cells is 3.8 points at $m{=}128$ (3.2 for the 8 cells that are not fully enumerated at $m{=}256$), but individual cells differ by up to 15.8 points (GMSC random forest: 79.0% vs. 94.8% at $m{=}128$; XGBoost 82.7% vs. 94.4%). The cause is the instance margin, not the noise: the median run-to-run SD of the attribution (relative to $\|\phi\|_2$) is 0.036 vs. 0.034 for GMSC-RF, but 54% of the explained instances of the first model have an exact gap $a_{(3)} - a_{(4)} < 0.02\|\phi\|_2$ versus 13% for the replicate. So absolute reliability levels in §4 are model-specific, which our instance-level intervals do not cover; the mechanism of §4.2 (reliability is set by gap-to-noise) is unchanged.

E Reproducibility and compute

Code, exact commands, versions and the mapping from tables/figures to scripts are in `README.md`; all numbers in this paper are produced by the scripts in `code/`. Total compute was about 10 CPU-hours for the final pipeline (about 3 more for aborted and exploratory runs; estimated as wall-clock times

Dataset	Model	top-3 set correct (%), m=128			top-3 set correct (%), m=256		
		main	replicate	AUC (m/r)	main	replicate	AUC (m/r)
pima	LR	98.4	97.6	0.867/0.814	enum.	enum.	0.867/0.814
pima	MLP	97.9	94.1	0.869/0.794	enum.	enum.	0.869/0.794
pima	RF	96.3	96.3	0.847/0.809	enum.	enum.	0.847/0.809
pima	XGB	91.5	94.4	0.834/0.790	enum.	enum.	0.834/0.790
gmsc	LR	95.1	95.8	0.853/0.853	98.3	97.0	0.853/0.853
gmsc	MLP	89.3	93.7	0.864/0.861	93.4	95.0	0.864/0.861
gmsc	RF	79.0	94.8	0.865/0.869	86.1	96.5	0.865/0.869
gmsc	XGB	82.7	94.4	0.866/0.872	88.3	96.8	0.866/0.872
acsincome	LR	94.1	95.7	0.889/0.889	96.5	97.4	0.889/0.889
acsincome	MLP	90.7	92.1	0.899/0.897	93.3	95.0	0.899/0.897
acsincome	RF	95.6	93.8	0.898/0.897	97.2	96.3	0.898/0.897
acsincome	XGB	94.4	94.0	0.902/0.901	96.8	97.1	0.902/0.901

Table 8: Main models (seed 0) vs. replicate models (split/model seed 1), Kernel SHAP, $d \leq 10$ datasets; AUC of both models given. "enum.": complete enumeration.

the 4-core quota; main experiment 4.9, background/retraining/pairing 1.0, library validation 0.7, stress test 1.8, replicate 1.1) CPU-hours on a machine whose cgroup quota is 4 cores. No dataset or result in this paper is synthetic; tabulated games are exact computations on the trained models.

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